Class Imbalance in Histopathology: Which Class Sits in the Tail Matters on BRACS, and Easy Classes Cost Most

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Abstract

On BRACS, a long-tailed training allocation at fixed patients and fixed patch budget costs 7.98 points of balanced accuracy at an imbalance ratio of $\rho = 100$, and the prior-versus-support study found that the two tail classes then keep only 7 percent recall. One reading of this large damage is a difficulty-by-prevalence interaction: hard, overlapping lesion classes might collapse when they are rare. This report tests how much the damage depends on which class sits in the tail. Per draw, the class order of the prevalence study is rotated through all seven cyclic shifts, so every class takes every rank exactly once, with patients, patches, and profile held fixed. The class in the tail changes the damage by up to 4.8 points at $\rho = 100$, from 5.76 points (95% interval: 3.95 to 7.58) with UDH in the tail to 10.55 points (8.82 to 12.35) with IC, and by up to 2.8 points at $\rho = 10$. The ordering is opposite to the overlap reading: damage rises with the recall a class has under balanced training (Spearman 0.96 at $\rho = 100$, seven classes, descriptive). In the tail at $\rho = 100$, six of seven classes keep at most 3 percent recall, so a class loses nearly all its balanced recall and easy classes lose the most. Only IC, the least confused class, keeps a substantial share, 44 percent of it. The class's own rank explains 74 to 81 percent of the variance of per-class recall changes, so the damage is largely a sum of per-class responses to rank. The BRACS damage therefore does not come from a few hard classes; it arises because almost any BRACS class collapses once it is rare.

1 Motivation

The BRACS prevalence study of this series measured how balanced accuracy on the seven BRACS lesion classes degrades when the training class distribution moves from balanced to long-tailed, with ten patients per class and a total of 2,240 patches held fixed (Queisler, 2026d). The damage reached 3.86 points at $\rho = 10$ and 7.98 points at $\rho = 100$, about seven times the damage of the same design on TCGA-UT (Queisler, 2026e). The prior-versus-support study split the BRACS damage at $\rho = 100$ into 4.47 points from the skewed class prior and 3.29 points from thin tail support. It also found that the two tail classes of r100 keep only 7.31 percent recall, against 38.72 percent under balanced training, and that a skewed prior alone lowers them to about 21 percent already at $\rho = 10$ (Queisler, 2026a).

In all of these studies the class order of the long-tail profile was drawn at random per draw, and the damage was averaged over the resulting assignments. That average hides a question the design did not answer: does the damage depend on *which* class is rare? Several BRACS lesion classes lie morphologically close to each other, and the balanced classifier recognizes them with very different recall. One reading of the large BRACS damage is therefore a difficulty-by-prevalence interaction: the damage would be large because hard, overlapping classes collapse when they are rare, and it would be largest when such a class sits in the tail.

Two mechanisms predict opposite orderings. Under the *overlap* mechanism, a class with much overlap loses its decision region first when its prior share shrinks, so hard classes collapse

faster and cost more in the tail. Under the *headroom* mechanism, the loss of a class in the tail is bounded by the recall it has under balanced training; if every class in the tail falls close to zero recall, easy classes have the most to lose and cost the most. This report asks: *how much does the damage at a fixed ratio depend on which class sits in the tail, and does the ordering follow overlap or headroom?*

2 Design

The study reuses the design of the BRACS prevalence curve (Queisler, 2026d): the seven BRACS lesion classes with frozen Virchow2 features (Zimmermann et al., 2024), three patient-disjoint splits with ten draws each, $G = 10$ patients per class, a total budget of $T = 2{,}240$ training patches, and nested patch orderings within each patient. The balanced arm r1 gives each class 320 patches. The ratio arm rρ gives the class at rank j a count proportional to $\rho^{-j/6}$, with rank 0 the head and rank 6 the tail; at $\rho = 100$ the seven ranks receive 1,206, 560, 260, 120, 56, 26, and 12 patches. Each draw assigns classes to ranks by a random permutation π.

Only the class-to-rank assignment varies here. For every split and draw, the shift-k arm assigns rank j to the class $\pi_{(j-k) \bmod 7}$, for $k = 0, \ldots, 6$ (Table 1). The seven shifts of one draw form a Latin square: every class takes every rank exactly once, and every class sits in the tail once per draw, 30 times per ratio. Patients, patch orderings, the per-rank counts, and the realized ratios (9.95 and 100.5) are identical across shifts. Shift 0 is the permutation of the prevalence study, so its fits and those of r1 are reused unchanged; the 6 other shifts at $\rho \in \{10, 100\}$ require $6 \times 2 \times 30 = 360$ fits, all of which converged. The reused shift-0 arms reproduce the stored balanced accuracies of the prevalence study exactly.

Rank j	0	1	2	3	4	5	6
Patches at $\rho = 100$	1,206	560	260	120	56	26	12
Shift 0 (prevalence study)	π_0	π_1	π_2	π_3	π_4	π_5	π_6
Shift 1	π_6	π_0	π_1	π_2	π_3	π_4	π_5
Shift 2	π_5	π_6	π_0	π_1	π_2	π_3	π_4
⋮							
Shift 6	π_1	π_2	π_3	π_4	π_5	π_6	π_0

Table 1: Cyclic shifts of a draw's class order place every class at every rank exactly once. π is the random class permutation of one split and draw in the prevalence study; shift k rotates this order by k ranks. The per-rank patch counts, patients, and patch orderings stay fixed, so only the identity of the class at each rank changes.

Cyclic shifts replace the random permutations of the prevalence study for two reasons. Thirty random permutations would place each class in the tail only about four times, whereas the Latin square places it there 30 times per ratio. And because the shifts keep the rest of the order fixed within a draw, the damage attributed to a tail class does not depend on a different random arrangement of the other six classes. The price is that, within a draw, a tail class always has the same neighbours at ranks 4 and 5; these neighbours vary across the 30 draws. Every fit selects its regularization strength by validation patient-macro balanced accuracy (Queisler, 2026c), as in the prevalence curve.

3 Evaluation

The primary endpoint is the *tail damage* of class c at ratio ρ,

$$D_\rho(c) = \mathrm{BA}(\mathrm{r1}) - \mathrm{BA}(\mathrm{r}\rho \mid c \text{ at rank } 6), \tag{1}$$

where BA is test patient-macro balanced accuracy and the second term pools the 30 fits in which c occupies the tail. The mean of $D_\rho(c)$ over the seven classes averages over all shifts and should match the prevalence damage up to the assignment noise of shift 0. Balanced accuracy is the mean of seven per-class recalls, so $D_\rho(c)$ splits exactly into the *own loss* $-\Delta_c/7$, where Δ_c is the change of the tail class's own recall against r1, and the *loss on the other classes*, $D_\rho(c)$ minus the own loss.

The spread of $D_\rho(c)$ over the seven classes is reported descriptively; its maximum minus minimum is biased upward by selecting the extremes of seven noisy means. The *rank response* $\Delta_\rho(c, j)$ is the change of the own recall of class c against r1 when c sits at rank j, a 7×7 matrix per ratio that the Latin square provides without additional fits. An *own-rank model* predicts every per-fit, per-class recall change by the mean of its (class, rank) cell; its R^2 bounds how much the arrangement of the other classes adds beyond each class's own rank.

Two class properties are computed from the balanced arm: the patient-macro recall of the class under r1, the *headroom*, and its *confusability*, the off-diagonal confusion mass into and out of the class as a share of all r1 test predictions, pooled over 30 fits. With seven classes, their Spearman correlations with $D_\rho(c)$ are descriptive, and no significance is claimed. The headroom mechanism predicts a positive correlation of $D_\rho(c)$ with r1 recall; the overlap mechanism predicts a positive correlation with confusability and a negative one with r1 recall.

Pooling and uncertainty follow the prevalence curve: equal weight per split, Bayesian bootstrap weights on test patients (Rubin, 1981) combined with resampling of draws within splits, 2,000 shared replicates, and 95 percent percentile intervals of every paired contrast. A difference counts as material if it exceeds one point of balanced accuracy, the threshold of the series. The ratios are $\rho = 10$ and $\rho = 100$, where the prevalence damage is 3.86 and 7.98 points.

4 Results

The results follow the endpoints: first the tail damage per class, then its split into own and other-class loss, the rank response with the additivity check, and finally the class properties. The balanced arm reaches 39.57 percent balanced accuracy (36.70 to 42.53), as in the prevalence curve. Averaged over the seven classes, the tail damage is 4.12 points (3.15 to 5.25) at $\rho = 10$ and 8.04 points (6.77 to 9.46) at $\rho = 100$, slightly above the 3.86 and 7.98 points of the prevalence study. The difference comes from the reused shift 0: at $\rho = 10$ the seven shifts reach between 34.73 and 36.03 percent balanced accuracy, and shift 0, at 35.71 percent, lies above their mean of 35.45; at $\rho = 100$ they reach 30.86 to 32.41 percent, and shift 0, at 31.59 percent, lies just above their mean of 31.53.

4.1 Tail damage per class

Which class sits in the tail changes the damage materially (Table 2, Figure 1). At $\rho = 100$ the damage ranges from 5.76 points with UDH in the tail and 5.88 points with ADH to 9.59 points with N and 10.55 points with IC, a spread of 4.8 points with a standard deviation of 1.68 points across classes. The intervals of the two extremes do not overlap. At $\rho = 10$ the damage ranges from 2.51 points with ADH to 5.31 points with FEA, a spread of 2.8 points with a standard deviation of 1.11 points. Both spreads exceed the one-point threshold, although, as stated in Section 3, the spread between the extremes of seven means overstates the true spread.

Tail class	r1 recall	Confusability	$D_{10}(c)$	$D_{100}(c)$
ADH	12.24	7.4	2.51 [1.14, 4.14]	5.88 [4.12, 7.89]
UDH	22.98	11.4	3.10 [1.70, 4.54]	5.76 [3.95, 7.58]
PB	31.42	23.7	3.02 [1.57, 4.63]	7.33 [5.26, 9.64]
DCIS	36.19	9.2	4.94 [3.43, 6.53]	8.30 [6.50, 10.17]
FEA	44.62	9.0	5.31 [3.67, 6.99]	8.85 [6.81, 11.01]
N	60.47	16.3	5.25 [3.72, 6.91]	9.59 [7.71, 11.66]
IC	69.08	6.3	4.73 [3.26, 6.27]	10.55 [8.82, 12.35]
Mean	39.57		4.12 [3.15, 5.25]	8.04 [6.77, 9.46]

Table 2: The damage grows with the balanced-arm recall of the class in the tail. Tail damage $D_\rho(c)$ is the balanced-accuracy loss against the balanced arm, in points with paired 95% bootstrap intervals, over the 30 fits in which class c occupies the tail. Classes are ordered by their patient-macro recall in the balanced arm, in percent; confusability is the off-diagonal confusion mass into and out of the class in percent of all balanced-arm test predictions.

The ordering does not follow the overlap reading, under which the hardest classes would cost the most. ADH and UDH, the two classes with the lowest balanced recall, are the cheapest tail classes at $\rho = 100$; at $\rho = 10$, ADH is the cheapest and UDH lies 0.08 points behind PB. Instead, the ordering follows the balanced-arm recall. The Spearman correlation between r1 recall and tail damage is 0.71 at $\rho = 10$ and 0.96 at $\rho = 100$, where only UDH and ADH swap places. The correlation with confusability is 0.00 and -0.29.

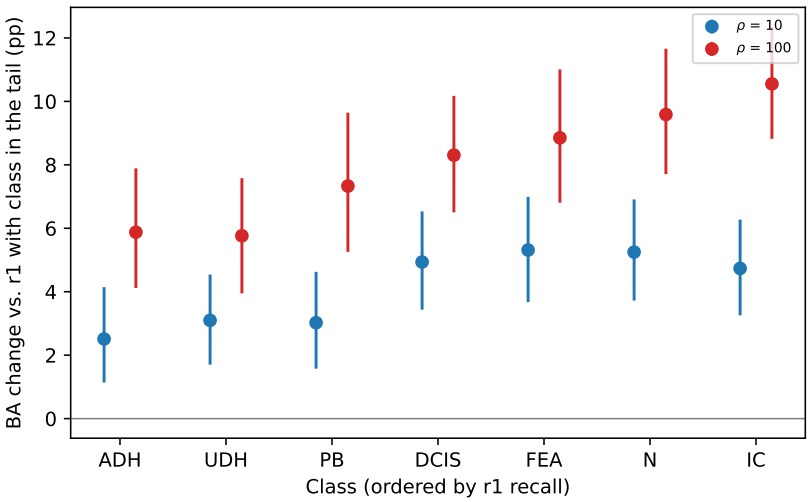


Figure 1: Easy classes cost the most when they sit in the tail. Balanced-accuracy loss against the balanced arm, in points, with the class on the horizontal axis placed in the tail, at $\rho = 10$ (blue) and $\rho = 100$ (red), with paired 95% bootstrap intervals. Classes are ordered by their recall in the balanced arm, from ADH (12 percent) to IC (69 percent).

4.2 Own loss and loss on the other classes

Most of the tail damage is the tail class's own lost recall (Table 3). At $\rho = 100$, six of the seven classes keep at most 3 percent recall in the tail: ADH 0.3, UDH 0.3, PB 2.4, DCIS 1.3, FEA 1.2, and N 2.6 percent. Each of them thus loses between 92 and 99 percent of its balanced recall,

and its own loss is close to its balanced recall divided by seven. Only IC keeps 30.6 percent, 44 percent of its 69.1 percent. At $\rho = 10$ the collapse is less complete and depends on the class: ADH keeps 1.6 percent, whereas FEA keeps 14.0, N 17.5, and IC 50.7 percent. Averaged over classes, the own loss is 3.47 of the 4.12 points at $\rho = 10$, 84 percent, and 4.86 of the 8.04 points at $\rho = 100$, 60 percent.

	Tail recall		Own loss			Loss on other classes		
Tail class	$\rho = 10$	$\rho = 100$	$\rho = 10$	$\rho = 100$		$\rho = 10$	$\rho = 100$	
ADH	1.6	0.3	1.52	1.71	[1.11, 2.48]	0.99	4.17	[2.50, 6.17]
UDH	5.3	0.3	2.52	3.24	[2.51, 4.05]	0.58	2.52	[0.60, 4.43]
PB	10.8	2.4	2.95	4.14	[3.08, 5.34]	0.07	3.19	[1.17, 5.51]
DCIS	7.1	1.3	4.16	4.98	[3.73, 6.22]	0.78	3.32	[1.71, 5.06]
FEA	14.0	1.2	4.37	6.20	[5.03, 7.54]	0.94	2.65	[1.17, 4.23]
N	17.5	2.6	6.14	8.27	[6.98, 9.48]	−0.89	1.32	[−0.63, 3.30]
IC	50.7	30.6	2.63	5.50	[4.56, 6.47]	2.11	5.06	[3.45, 6.68]

Table 3: In the tail at $\rho = 100$ every class except IC loses almost all of its recall, so its own loss tracks its balanced recall. Tail recall is the mean own patient-macro recall of the class in percent when it occupies the tail. The own loss is the drop of that recall against the balanced arm divided by seven, the share of balanced accuracy it removes; the loss on the other classes is the rest of $D_\rho(c)$. Both are in points, with paired 95% bootstrap intervals at $\rho = 100$; the $\rho = 10$ intervals of the other-class loss contain zero for every class except IC (0.71 to 3.57).

The loss on the other classes is small at $\rho = 10$, where its interval excludes zero only with IC in the tail, and it grows to between 1.32 and 5.06 points at $\rho = 100$. It is largest with IC and ADH in the tail and smallest with N. For ADH, it carries 71 percent of the damage. ADH has little recall of its own to lose, but it is also among the classes that gain most at the upper ranks (Section 4.3); with ADH in the tail, those ranks go to classes that gain less there. Because the arrangement of the other six classes is tied to the tail class within each draw (Section 2), these per-class values of the other-class loss are not interpreted further.

4.3 Rank response and additivity

Every class responds to its rank in the same monotone way, and the size of the response follows its headroom in both directions (Figure 2). At $\rho = 100$ a class at the head gains between 8.7 points of recall (IC) and 41.4 points (UDH); the classes with low balanced recall, UDH, ADH, and PB, gain the most, by 32 to 41 points, while N and IC, which have the least room above them, gain 15.9 and 8.7 points. Toward the tail, the order reverses: N loses 57.9 points and FEA 43.4, while ADH and UDH lose 11.9 and 22.7 points. For the low-recall classes the loss saturates before the tail: ADH loses 11.4 points at rank 5 and 11.9 at rank 6, and it has almost nothing left to lose beyond. The step from rank 5 to rank 6, where the class count falls from 26 to 12 patches, removes a further 4.4 to 9.4 points from DCIS, FEA, PB, N, and IC. At rank 3, with 120 patches, the recall of ADH, UDH, and PB is within 5 points of the balanced arm, whereas N already loses 24.9 points and FEA 18.8.

The own-rank model explains 74 percent of the variance of the per-fit, per-class recall changes at $\rho = 10$ and 81 percent at $\rho = 100$. The remaining 26 and 19 percent contain the effect of which other classes hold the other ranks, but also the fit-to-fit variation of per-class recall over test patients and draws, which the design cannot separate from it. The pairings of classes can therefore explain at most about a quarter of the variance, and the damage is largely the sum of the separate responses of the seven classes to their own ranks.

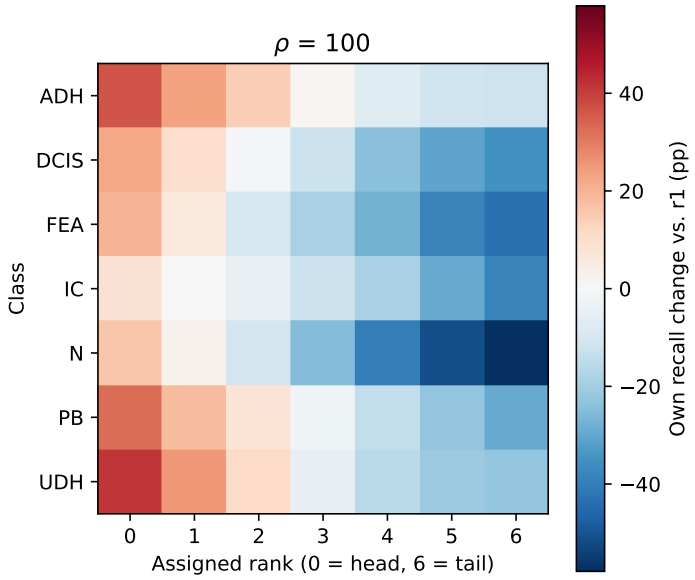


Figure 2: Every class gains recall at the head and loses it toward the tail, and the easy classes N and IC gain least and lose most. Change of each class's own patient-macro recall against the balanced arm, in points, when the class sits at the given rank at $\rho = 100$; rank 0 receives 1,206 patches and rank 6 receives 12. Each cell pools the 30 fits in which that class holds that rank.

4.4 Class properties

Of the two class properties, only the headroom orders the damage. The own loss of the tail class correlates with its balanced recall at 0.64 at $\rho = 10$ and 0.89 at $\rho = 100$, and with confusability at 0.39 and 0.07. Confusability does not follow balanced recall: PB and N are the most confused classes of the balanced arm, with 23.7 and 16.3 percent of all predictions, while ADH, the class with the lowest recall, lies at 7.4 percent. The least confused class, IC, is the only one that keeps a substantial share of its recall in the tail.

In relative terms, harder classes do collapse faster, as the overlap mechanism predicts, but only at the moderate ratio. At $\rho = 10$, ADH loses 87 percent of its balanced recall, DCIS 80 and UDH 77 percent, while FEA, N, and PB lose 66 to 71 percent and IC 27 percent; the Spearman correlation between balanced recall and the lost fraction, computed from Table 3, is -0.68. At $\rho = 100$ all classes except IC lose 92 to 99 percent, and the lost fraction no longer distinguishes them.

5 Discussion

The class in the tail matters on BRACS: at $\rho = 100$ it moves the damage between about 5.8 and 10.6 points, a range wider than the whole prevalence damage at $\rho = 10$. The ordering is not the one the difficulty-by-prevalence reading predicts. The hardest classes are the cheapest in the tail, not the most costly. The ordering follows the headroom mechanism instead: at $\rho = 100$ nearly every class loses almost all of its recall when it is rare, so the damage a class causes in the tail is set by how much recall it had to lose. IC and N, the classes the balanced classifier recognizes best, cause the largest damage; ADH, which the balanced classifier recognizes in only 12 percent of patients, causes one of the smallest.

The overlap mechanism is not refuted but acts on a different scale. At $\rho = 10$ the harder classes lose a larger fraction of their recall than the easy ones, and IC, the most distinct class of

the balanced arm, is the only class that survives the tail at $\rho = 100$ with a substantial share, 44 percent, of its recall. Overlap thus decides how fast and how completely a rare class collapses, and headroom decides what this collapse costs in balanced accuracy. On BRACS the collapse is nearly complete for six of seven classes at $\rho = 100$, so headroom dominates the damage.

This refines the account of the BRACS damage given by the preceding studies. The prevalence study found a damage seven times that of TCGA-UT (Queisler, 2026d), and the prior-versus-support study found that the tail classes fall to a recall floor that further imbalance hardly moves (Queisler, 2026a). The present results show that this floor is close to zero for six of the seven classes and not specific to the hard ones. The large BRACS damage therefore does not come from one hard class that happens to be rare; it arises because almost any BRACS class disappears from the predictions once it is rare, whereas the tail third on TCGA-UT keeps about 60 percent recall at $\rho = 100$ (Queisler, 2026b). The high sensitivity of BRACS to imbalance is a property of the dataset as a whole under ten patients per class, not of a few hard classes.

For the design of imbalance studies, the results support the random class order per draw used throughout this series. A study that fixes one class order on BRACS would report a damage anywhere between about 5.8 and 10.6 points at $\rho = 100$, depending on which class it happened to make rare, and, with R^2 values of 0.74 and 0.81 for the own-rank model, most of this variation is predictable from the rank response of each class. Averaging over assignments, as the prevalence study does, estimates the damage for an unknown rare class; the tail damage per class answers the more specific clinical question of what happens when a given lesion type is the rare one.

6 Limitations

With seven classes, the correlations with class properties rest on seven points and are descriptive; the per-class damages have intervals of about ± 2 points, so only the extremes of the ordering are well separated. The Latin square is built from cyclic shifts of one random order per draw, so within a draw the tail class always has the same neighbours at the next ranks; these vary across the 30 draws but are not balanced, and the other-class loss per tail class depends on them. The own-rank model cannot separate the effect of class pairings from fit-to-fit variation, so its R^2 gives only a lower bound on additivity.

The confusability measure pools patch-level confusion counts and therefore gives classes with many test patches more weight, and it does not capture overlap as a pathologist would rate it. The study covers two ratios and one exponential profile. All results are conditional on ten patients per class, frozen Virchow2 features, multinomial logistic regression, BRACS, and three overlapping splits with about 24 test patients each.

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